

A PCA–VAR Study of Predictive Economic Linkages among Guangdong Cities

Ruikang Ding[†]

Dian Jin[†]

School of Mathematical Sciences, Fudan University, Shanghai, China

[†]Equal contribution.

Abstract

This study uses annual GDP series for 21 Guangdong cities from 1999 to 2023 to construct a directed network of cross-city predictive linkages. The studyflow combines stationarity preprocessing, principal-component analysis (PCA), vector autoregression (VAR), pairwise Granger tests, centrality measures, and impulse-response analysis. With the project’s threshold rule $F > 2.5$, the resulting network contains 336 directed edges, an average total degree of 32, and density 0.80. Degree and eigenvector centrality reveal substantial heterogeneity in network position, while the impulse-response matrix contains both positive and negative responses. We distinguish Granger predictability from structural causality and treat the $F > 2.5$ rule as an exploratory edge threshold rather than a conventional significance statement.

Keywords: regional economic network; Guangdong; PCA; VAR; Granger predictability; centrality

1 Introduction

Economic activity across neighboring cities is not independent. Trade, commuting, industrial supply chains, investment, infrastructure, and common macroeconomic conditions can make past changes in one city informative about later changes elsewhere. A directed network offers a compact way to represent such asymmetric predictive relationships.

The source project studies Guangdong using annual GDP for 21 cities over 1999–2023. Its analysis proceeds from time-series preprocessing to PCA, VAR estimation, Granger tests, and network statistics. That sequence is retained here, but the interpretation is tightened. A Granger relation is a statement about incremental predictive information in a specified dynamic model; it is not, by itself, evidence that one city’s GDP structurally causes another city’s GDP. Granger’s original formulation makes this forecasting interpretation explicit.[1]

Network representations of dynamic dependence are well established in empirical economics. For example, Diebold and Yilmaz show how directed weighted networks can be constructed from variance-decomposition measures to study connectedness.[2] The present project is simpler and more exploratory: it uses pairwise Granger F statistics as edge weights and then examines the resulting topology.

2 Data and Methods

2.1 Data and Preprocessing

The dataset described in the analysis consists of annual GDP observations for 21 Guangdong cities from 1999 through 2023. The analysis first performs stationarity checks and differencing as needed before fitting a VAR. It then applies PCA to the multivariate series to extract dominant common variation and reduce dimensionality.

This reduction is practically important. With only 25 annual observations per city, a large unrestricted VAR would have many parameters relative to the sample size. PCA lowers the dimension of the dynamic system, although it also changes interpretation: principal components are linear combinations of the original city series.

PCA–VAR and Granger Testing. Let Y_t denote the vector of retained dynamic variables after preprocessing. A VAR of order p has the form

$$Y_t = c + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + u_t. \quad (1)$$

For a city pair $X \rightarrow Y$, the Granger procedure compares a restricted predictive equation for Y with an unrestricted equation that includes lagged information associated with X . If those additional terms improve prediction sufficiently, the project records a directed link from X to Y .

The resulting F statistics form a 21×21 weighted matrix. Directed links with $F > 2.5$ are retained. We refer to these edges as *thresholded Granger-predictive links* rather than treating the threshold alone as a formal causal-significance

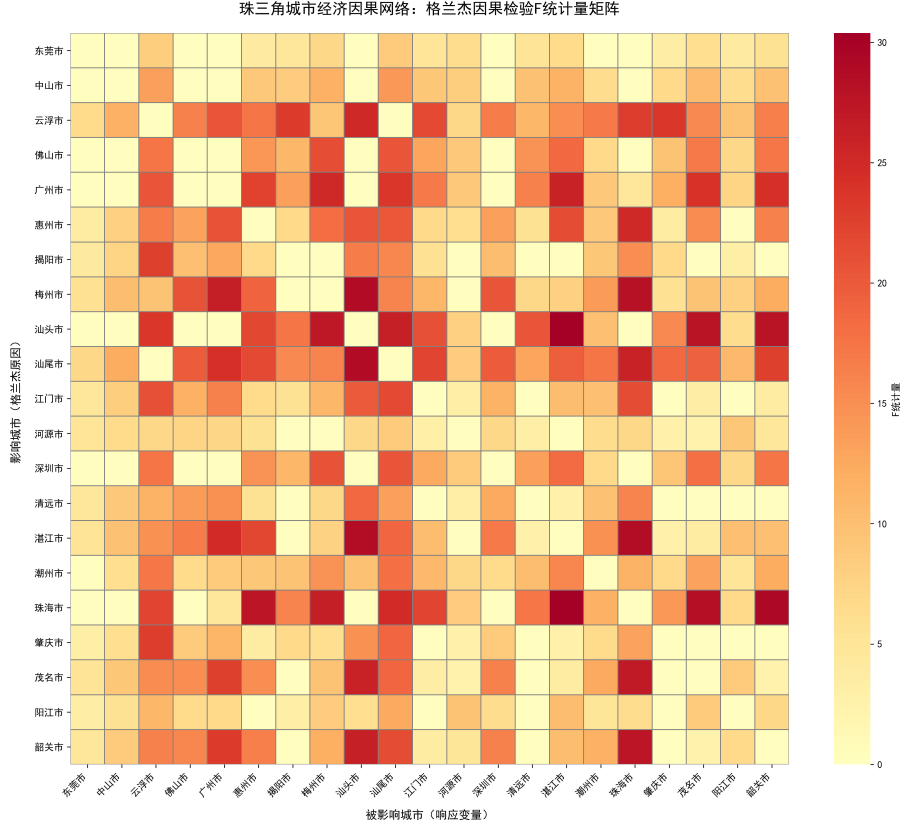


Figure 1: Granger F -statistic matrix for the 21-city analysis.

Table 1: Network statistics for the thresholded directed graph.

Statistic	Value
Nodes	21
Directed edges	336
Average total degree	32.00
Density	0.8000
Highest composite centrality (report)	Dongguan

criterion. An F statistic becomes a formal significance statement only after its degrees of freedom and reference distribution are specified; multiple testing should also be considered when interpreting a large set of pairwise tests.

Network Construction. The network has 21 nodes, one for each city. A directed edge represents a retained Granger-predictive relation, and the corresponding F statistic is used as its weight. Under the $F > 2.5$ rule, the source summary reports 336 directed edges, average total degree 32.00, and network density 0.8000.

A density of 0.80 means that the chosen threshold leaves a very dense graph. This is a useful descriptive result, but it also means that conclusions about a small set of “core” nodes should be tested for sensitivity to the threshold. If most possible links survive, a modest change in cutoff can alter degree rankings and the apparent core-periphery structure.

3 Results

3.1 Centrality

The project considers degree centrality and eigenvector centrality. Degree centrality records how many direct links a node has, whereas eigenvector centrality gives greater weight to links involving nodes that are themselves central. These measures answer different questions and need not rank cities identically.

The source text highlights Dongguan and Zhongshan as cities with comparatively high centrality, and the appendix identifies Dongguan as highest in the composite centrality score. Separately, the visualization summary reports Yunfu as having the highest in-degree, out-degree, and total degree. These statements are not necessarily contradictory because

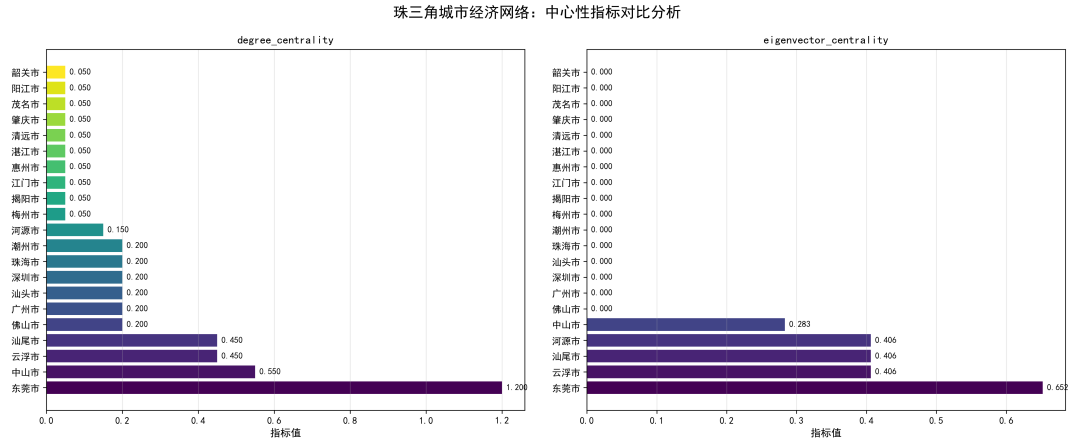


Figure 2: Degree and eigenvector centrality for the 21-city network.

Table 2: Partial matrix reported in the original appendix.

	Dongguan	Zhongshan	Yunfu	Shanwei	Heyuan
Dongguan	0.000	5.413	-5.369	-5.421	-5.093
Zhongshan	4.195	0.000	3.956	2.043	2.946
Yunfu	-0.530	-0.412	0.000	-0.396	0.850
Shanwei	6.280	4.126	-3.826	0.000	3.140
Heyuan	3.052	0.850	-1.180	-1.698	0.000

they refer to different statistics, but they should not be collapsed into a single ranking.

The centrality plot also shows how concentrated the eigenvector measure is: most cities have values near zero, while a small group carries the nonzero mass shown in the figure. In the degree panel, Dongguan has the largest displayed normalized value, followed by Zhongshan and two cities at 0.45. These are properties of this thresholded predictive network, not a general ranking of economic importance.

Impulse Responses. The analysis complements the Granger network with an impulse-response (IRF) matrix. An IRF traces the fitted response of the system after an innovation to one variable. The analysis interprets positive responses as possible spillovers and negative responses as possible competitive or siphoning effects. Those economic labels are plausible hypotheses, but a reduced-form innovation is not automatically a structural economic shock.

The network visualization is described as exhibiting high connectivity and a core-periphery pattern. The project further compares in-degree, out-degree, and total degree, and forms a composite score by standardizing degree and eigenvector centrality. These visualizations are useful summaries of the fitted model, especially because directionality is difficult to see in the raw 21×21 matrix.

Reported Matrix Excerpt. A small numerical block of the Granger matrix is shown below to illustrate the scale and directionality of the pairwise statistics.

The presence of negative entries in this table deserves attention because a conventional Granger-test F statistic is nonnegative. Accordingly, the displayed matrix is interpreted conservatively as the pairwise score matrix used in the network-construction stage rather than as a set of formally calibrated F statistics. The thresholded-network conclusions below rely on the stated edge-selection rule and are not used to infer structural causality.

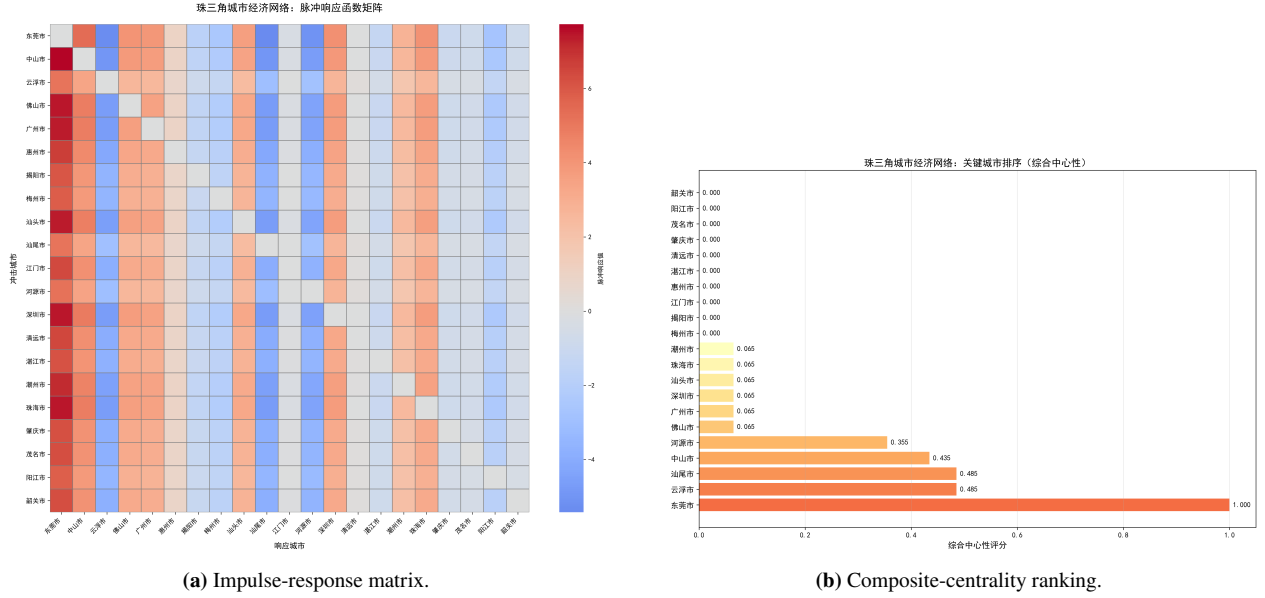


Figure 3: Additional results from the city-network analysis.

Directed-Network Construction from the F Matrix. The 21×21 pairwise matrix provides the bridge between time-series modeling and network analysis. This step determines the interpretation of every subsequent graph statistic. For each ordered pair (i, j) , a scalar test output is assigned to the directed relation $i \rightarrow j$. The diagonal is set to zero. After thresholding at 2.5, the surviving ordered pairs become edges; the stored scalar is then used as the edge weight.

There are $21 \times 20 = 420$ possible off-diagonal directed links. The reported 336 retained links therefore give $336/420 = 0.80$, exactly matching the stated network density. The reported average total degree of 32 is also internally consistent: in a directed graph, the sum of in-degree plus out-degree over all nodes equals twice the number of edges, so $2 \times 336/21 = 32$.

This arithmetic check is useful because it verifies that the headline topology statistics refer to the same thresholded graph. It does not resolve the earlier issue that some entries in the appendix matrix are negative despite being labeled as F statistics. For that reason, the present paper preserves the project’s topology but does not reinterpret the signed cells as conventional F values.

Centrality and Degree Structure. Degree-based statistics separate two roles that are combined in total degree. In-degree counts the number of retained links entering a city and therefore summarizes how broadly its series is predicted by lagged information elsewhere under the fitted model. Out-degree counts outgoing links and summarizes how broadly its lagged information contributes to prediction elsewhere. Total degree is their sum.

The analysis uses these quantities together with eigenvector centrality. If A denotes the adjacency matrix, eigenvector centrality is based on a relation of the form

$$\mathbf{x} = \lambda^{-1} A \mathbf{x}, \quad (2)$$

so a node receives a high score when it is linked to other high-scoring nodes. This differs from degree, which counts links without recursively weighting the importance of neighbors. The distinction explains why the project can identify Dongguan as highest in its composite centrality while the visualization summary separately identifies Yunfu as leading the raw degree counts.

The interpretation distinguishes three cases: cities high on both degree and eigenvector centrality, cities with high degree but low eigenvector centrality, and cities with low degree but high eigenvector centrality. The first group is broadly connected and embedded among central neighbors. The second has many direct links but is connected to less central nodes. The third has fewer direct links but connects to structurally important parts of the graph. This classification is more informative than treating any one centrality statistic as an overall economic ranking.

Impulse Responses as a Complementary Dynamic View. The Granger network and the impulse-response matrix answer related but different questions. A retained Granger link says that lagged information improves prediction in the specified model. An impulse response asks how the fitted system evolves after an innovation. The source project uses the latter to distinguish positive and negative dynamic responses that are not visible from edge existence alone.

In a reduced-form VAR, however, the meaning of a “one-standard-deviation shock” depends on how innovations are identified. The interpretation of a “one-standard-deviation shock depends on the innovation-identification convention.

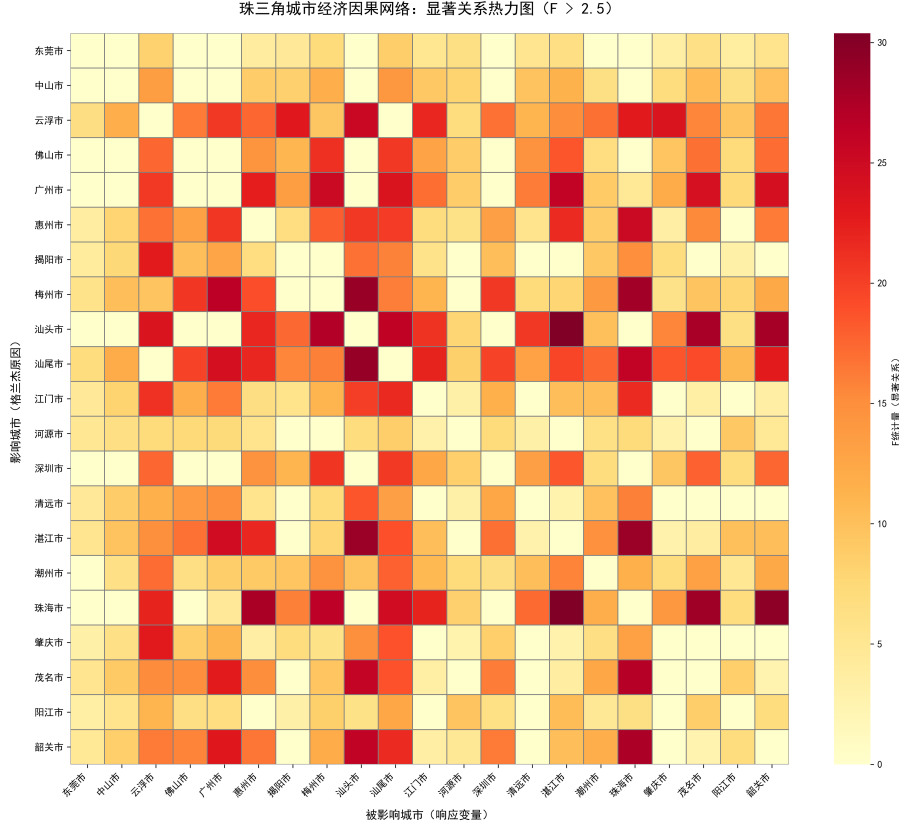


Figure 4: Thresholded Granger-predictive relation matrix from the original analysis, retaining entries with $F > 2.5$. The threshold is treated here as an exploratory edge-retention rule rather than a universal significance cutoff.

We therefore emphasize the qualitative observation that both positive and negative responses occur and avoid structural language about policy shocks or direct transmission channels.

The project interprets positive responses as possible regional synergy and negative responses as possible competition or siphoning. Those labels are useful economic hypotheses for follow-up work. A stronger design would pair the time-series evidence with observable mechanisms such as intercity freight, firm ownership links, commuting, sector composition, or infrastructure changes.

4 Discussion

4.1 Threshold Choice and Robustness

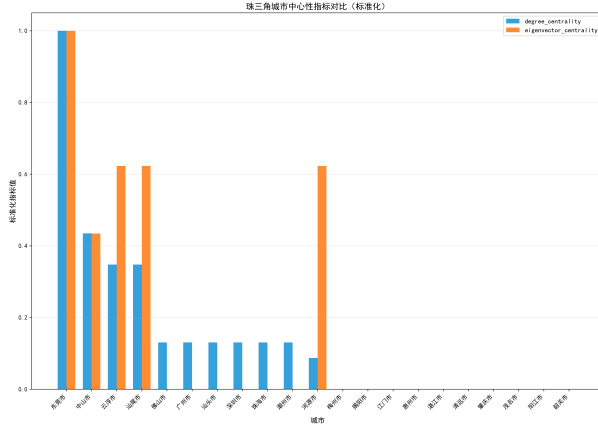
The threshold $F > 2.5$ is consequential. At the reported cutoff, four-fifths of all possible directed links remain. A dense graph can be a genuine feature of a highly integrated regional economy, but it can also arise when common trends, shared province-wide shocks, or a permissive threshold generate many pairwise predictive relations.

A direct robustness analysis would rebuild the graph over a sequence of thresholds and record density, the largest strongly connected component, degree rankings, and rank correlations of centrality. If Dongguan, Zhongshan, Yunfu, and other highlighted cities remain prominent over a broad range, the core-periphery interpretation would be more persuasive. If rankings change sharply, the appropriate conclusion would be that node importance is threshold-dependent.

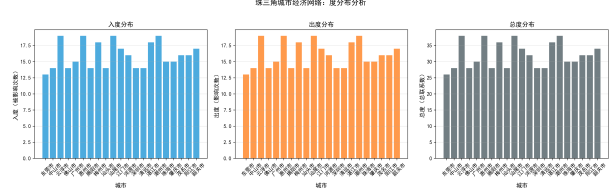
The second issue is multiple testing. With 420 ordered off-diagonal pairs, testing each pair separately can create many false discoveries even when individual tests use conventional levels. A more formal inferential analysis should report p-values, degrees of freedom, and a false-discovery or other multiple-testing adjustment alongside the raw test statistic.

The third issue is sample length. Twenty-five annual observations are few for estimating rich multivariate dynamics, especially after differencing and lagging. PCA reduces the parameter burden, but it does not create additional temporal information. Quarterly data or a broader panel of economic indicators would allow a more stable examination of changing network structure.

Economic Interpretation within the Evidence. Three descriptive features stand out in the fitted graph: high connectivity, asymmetry, and a core-periphery pattern. Their economic meaning should nevertheless remain conditional



(a) Normalized centrality comparison.



(b) In-, out-, and total-degree distributions.

Figure 5: Network diagnostics, including degree distributions and the composite key-city ranking.

on the model. An outgoing predictive link may reflect direct industrial transmission, but it may also reflect a common omitted driver that reaches one city earlier than another. Likewise, a central city in this graph is central to the selected GDP-predictability representation, not necessarily to every dimension of regional integration.

This distinction does not make the network uninformative. On the contrary, it suggests a practical use: the graph can be treated as a screening device for city pairs and clusters that deserve mechanism-specific investigation. The network reduces a large set of pairwise dynamic relations to a tractable map; subsequent economic analysis can then ask which links correspond to trade, investment, labor mobility, or shared sectoral exposure.

Robustness and Limitations. Three checks would materially strengthen the analysis. First, the network should be rebuilt across a grid of edge thresholds and VAR lag orders. Stability of the centrality ranking would show whether the identified core is a robust feature rather than a cutoff artifact. Second, the 420 ordered off-diagonal city pairs create a multiple-testing problem. A formal analysis should report p-values and control family-wise error or false discovery rate. Third, the 1999–2023 interval spans substantial structural change in Guangdong; a single fixed network assumes stability over the entire period.

The annual frequency is itself restrictive. Twenty-five observations leave little room for rolling estimation or high-order dynamics. Quarterly output, industrial production, freight flows, electricity consumption, firm-level supply links, or other higher-frequency indicators would make a time-varying network more defensible.

These caveats do not make the exercise uninformative. The project successfully turns a high-dimensional set of city series into a directed descriptive network and uses several network statistics to expose heterogeneity that is difficult to see from individual GDP trajectories. The appropriate conclusion is about predictive connectedness under the chosen specification, not structural causal transmission.

Visual Topology and Key-City Screening. The directed-network rendering complements the centrality plots. It uses node size to represent activity, arrow direction for the retained predictive relation, and edge thickness for the reported relation strength. At the $F > 2.5$ cutoff the graph is visually dense, which is consistent with the numerical density of 0.80 rather than an artifact of layout alone.

The topology exhibits a core–periphery structure, with Dongguan prominent in several centrality views. The centrality measures need not agree because they encode different network roles. A practical reading is therefore to use the composite ranking as a screening device rather than to assign a single definitive ordering to all cities. The strongest candidates can then be examined with the underlying time series, sector composition, and pairwise dynamic results.

Regional Interpretation. The analysis motivates the network in four ways. First, it turns an abstract collection of cross-city relations into quantities that can be compared. Second, the graph and heat maps make asymmetry visible: a city can be highly predictive of others without being equally predictable from them. Third, the IRF layer adds a dynamic view of how disturbances propagate through the fitted system. Fourth, the combined representation can help identify city pairs or clusters for more detailed regional analysis.

These uses remain defensible as exploratory analysis. Direct policy intervention, risk prevention, or identification of growth-driving cities would require stronger identification than the present Granger-network analysis provides. In particular, a dense Granger network alone cannot establish that intervention in a central city will mechanically transmit to its neighbors.

A more evidence-matched application is monitoring. If a city has high out-degree or sits among high-eigenvector

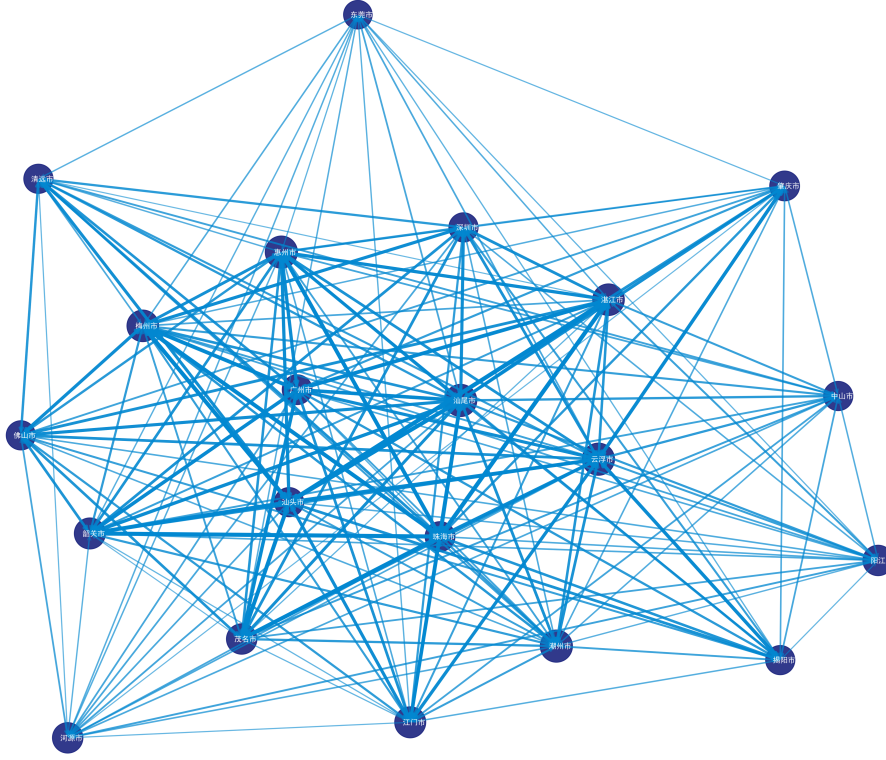


Figure 6: Directed network of Granger-predictive relationships among the 21 Guangdong cities after thresholding the pairwise test matrix.

nodes, an unusual movement in its economic indicators may justify closer examination of connected cities in the fitted model. This is a forecasting and surveillance use of the network, consistent with the predictive nature of Granger analysis.

5 Conclusion

The PCA–VAR workflow produces a dense directed network of Guangdong city GDP linkages, with 336 retained edges and density 0.80 under the project’s $F > 2.5$ rule. Centrality and impulse-response results indicate heterogeneous network positions and asymmetric dynamic relationships. At the same time, the short annual sample and the exploratory thresholding rule limit causal interpretation. The natural next step is not a more elaborate visualization but a more explicit statistical specification and robustness analysis.

References

- [1] C. W. J. Granger, “Investigating causal relations by econometric models and cross-spectral methods,” *Econometrica*, vol. 37, no. 3, pp. 424–438, 1969.
- [2] F. X. Diebold and K. Yilmaz, “On the network topology of variance decompositions: Measuring the connectedness of financial firms,” *Journal of Econometrics*, vol. 182, no. 1, pp. 119–134, 2014, doi:10.1016/j.jeconom.2014.04.012.
- [3] M. Billio, M. Getmansky, A. W. Lo, and L. Pelizzon, “Econometric measures of connectedness and systemic risk in the finance and insurance sectors,” *Journal of Financial Economics*, vol. 104, no. 3, pp. 535–559, 2012.